# Dr. Ponnappa Kechanda Prasanna

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## Education

June 21 - Jan 26
**Ph. D. - Physics**
MATES LAB (Supervisor : Dr. Sudip Chakraborty)
Harish-Chandra Research Institute (HRI) Prayajgraj, Uttar Pradesh (India) - 211019
(An aided instute of the DAE, A C.I. of Homi Bhabha National Institute )
Thesis title: *Theoretical Study of Lower Dimensional Materials for Energy Scavenging.*

Oct 19 – May 21
**Ph. D. - Physics** (Group moved to HRI)
MATES LAB (Supervisor : Dr. Sudip Chakraborty)
Indian Institute of Technology, Indore (IITI), Madhya Pradesh (India)-452020
Thesis title: *Theoretical Study of Lower Dimensional Materials for Energy Scavenging.*

Aug 16 - May 18
**M.Sc. - Physics (7.526 CGPA)**
Specialization: *Condensed Matter Physics.*
Manasagangotri - Mysore, Karnataka (India)-570006
University of Mysore
Thesis title: *Simulation of Diffraction from Rectangular Apertures*
(Supervisor: M. A. Shridhar)

Aug 13 - May 16
**B.Sc. (79.58%)**
Specialization: *Physics, Mathematics, Computer Science.*
SBRR Mahajana FGC, Mysore, Karnataka (India) - 570012
University of Mysore

## Competative Exam / Fellowship

Infosys Fellowship — Aug 2024 - Aug 2025

JOINT CSIR - UGC NET EXAMINATION — **AIR 94** (June 2019)
Physics

GATE — **AIR 382, SCORE 559** (March 2019)
Physics

JEST — **AIR 550, 92.46 percentile** (March 2019)
Physics

KSET — March 2019
Physics

## Teaching Experience

Jan 24 - May 24
**Teaching Assistant** HRI
Tutor for Advanced Numerical Analysis Course for First year M.Sc and Ph.D

Nov 19
**Teaching Assistant** IITI
Tutor for Advanced Numerical Analysis Course for Second year M.Sc

Aug 19 - Sep 19
**Guest Lecturer,** GFGC Kuvempunagar, Mysore, Karnataka (India) - 570023
Handled first- and second-year lab experiments, teaching Modern Physics to first-year B.Sc. students and Electronics to second-year B.Sc. students.

## Research Interests

Density Functional Theory(DFT), Two Dimensional Materials, Photovoltaics, Photoelectrocatalysis, Excitons

## Skills

| | |
|---|---|
| Languages | Reading, writing and speaking ability for English, Kannada, Hindi & Kodava Thakk. |
| Coding | $C$, $C^{++}$, Fortran, Python, Perl, GNU Octave, |
| First-principle calculations | Vienna Ab initio Simulation Package(VASP) and Quantum Espresso (QE) and Yambo |

## Research Publications

### Journal Articles


1. **P. K. Prasanna**, S. Bhattacharya, and S. Chakraborty, "Interfacial configuration impact on rashba spin splitting, excitonic radiative lifetime and band edge alignment in janus monolayers based heterostructures," *Journal of Materials Chemistry A*, vol. 13, pp. 32 232–32 245, 2025. DOI: https:/doi.org/10.1039/D5TA04076K.
2. **P. K. Prasanna** and S. Chakraborty, "Emerging chalcogenide based van der waals heterostructures for ultrathin excitonic solar cells with enhanced photo-conversion efficiency," *Journal of Materials Chemistry A*, vol. 13, pp. 7949–7961, 2025. DOI: https:/doi.org/10.1039/D4TA08172B.
3. **P. K. Prasanna** and S. Chakraborty, "Threefold impact on reaction coordinate mapping of zirconium based tri-chalcogenide pseudo-monolayers as emerging ultrathin catalysts for water splitting," *Advanced Theory and Simulations*, vol. 8, p. 2 400 816, 2024. DOI: https:/doi.org/10.1002/adts.202400816.
4. R. Sadhukhan, A. Kumar, **P. K. Prasanna**, A. Guha, R. Arenal, S. Chakraborty, and T. N. Narayanan, "Ultra-low-loaded platinum bonded hexagonal boron nitride as stable electrocatalyst for hydrogen generation," *ACS Applied Materials & Interfaces*, vol. 16, no. 13, pp. 17 063–17 064, 2024. DOI: https:/doi.org/10.1021/acsami.4c04036.
5. D. Singh, S. Singh, **P. K. Prasanna**, R. K. Rai, P. Chirawatkul, S. Chakraborty, M. Fichtner, and P. Barpanda, "Zinc-substituted cobalt phosphate ZnCo2(PO4)2 as a bifunctional electrocatalyst," *Inorganic Chemistry*, vol. 62, no. 31, pp. 12 345–12 355, 2023. DOI: https:/10.1021/acs.inorgchem.3c01367.
6. P. F. M. Elango, S. Kabir, **P. K. Prasanna**, M. X. Low, M. Yang, S. Chakraborty, S. Walia, S. Sriram, and M. Bhaskaran, "Vanadium dioxide-based miniaturized thermal sensors: Humidity effects on phase change and sensitivity," *ACS Applied Electronic Materials*, vol. 4, no. 11, pp. 5456–5467, 2022. DOI: https:/doi.org/10.1021/acsaelm.2c01110.
7. S. Mondal, S. Sarkar, D. Bagchi, T. Das, R. Das, A. K. Singh, **P. K. Prasanna**, C. Vinod, S. Chakraborty, and S. C. Peter, "Morphology-tuned Pt3Ge accelerates water dissociation to industrial-standard hydrogen production over a wide ph range," *Advanced Materials*, vol. 34, no. 30, p. 2 202 294, 2022. DOI: https:/doi.org/10.1002/adma.202202294.
8. A. G. Saraswathy Vilasam, **P. K. Prasanna**, X. Yuan, Z. Azimi, F. Kremer, C. Jagadish, S. Chakraborty, and H. H. Tan, "Epitaxial growth of gaas nanowires on synthetic mica by metal–organic chemical vapor deposition," *ACS Applied Materials & Interfaces*, vol. 14, no. 2, pp. 3395–3403, 2022. DOI: https:/doi.org/10.1021/acsami.1c19236.

9. N. Sethulakshmi, S. Nellaiappan, **P. K. Prasanna**, T. Das, S. Irusta, S. Chakraborty, and S. Sharma, "Nanocoral architecture for enhanced hydrazine assisted water oxidation: Insight from experiment and theory," *Journal of Electroanalytical Chemistry*, vol. 922, p. 116 776, 2022. DOI: https:/doi.org/10.1016/j.jelechem.2022.116776.

10. S. Parveen, **P. K. Prasanna**, S. Chakraborty, and P. Giri, "Stable deep blue emission with unity quantum yield in organic–inorganic halide perovskite 2d nanosheets doped with cerium and terbium at high concentrations," *Journal of Materials Chemistry C*, vol. 9, no. 7, pp. 2437–2454, 2021. DOI: https:/doi.org/10.1039/D0TC04937A.

## Under Revision/To be Submitted

1. **P. K. Prasanna** and S. Chakraborty, "Many-body pertubration theory based probing of inter and intra-layer exciton delocalization in indium-based chalcogenide heterostructures,"

## Workshops & Conferences

15-26 Jul 2024 — **Poster Presentation** "Emerging Van der Waals Heterostructures for Ultrathin Solar Cells: Implication of Type-II Band Edge Alignment on Photo-Conversion Efficiency"
**Engineered 2D Quantum Materials**
ICTS Banglore, India

22-26 Apr 2024 — **Oral presentation** "Emerging Van der Waals Heterostructures for Ultrathin Solar Cells: Implication of Type-II Band Edge Alignment on Photo-Conversion Efficiency"
**2024 MRS spring meeting & exhibit**
Seattle, Washington

12-15 Dec 2023 — **Poster Presentation** "Emerging Van der Waals Heterostructures for Ultrathin Excitonic Solar Cells"
**34th AGM of MRSI and 5th Indian Materials conclave**
IIT BHU, India

24-28 July 2023 — **Poster Presentation** "Enhancing the Catalytic Activity of Zirconium Tri-Sulfides by Functionalization, Vacancy defect and Strain Engineering"
**NAMMA Psi-k Workshop**
JNCASR & IISC Banglore, India

2-4 Feb 2023 — **Poster Presentation(Best Poster)** "Enhancing the Catalytic Activity of Zirconium Tri-Sulfides by Functionalization, Vacancy defect and Strain Engineering"
**Conference on Advances in Renewable Energy**
HRI & TIFR Hyderabad, India

## References

**Prof. Sudip Chakraborty**
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Harish-Chandra Research Institute,
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**Prof. Chennupati Jagadish**
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